Advanced Programming Practice · 21CSC203P

Unit 4 · Session 11 · SLO 2 — Solved Worksheet

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Unit â€“ 4 : Session â€“ 11 : SLO â€“ 2 | Converting a TCP Client to UDP

■ I. Key Changes (TCP client â†’ UDP client)

1. Socket type: socket.SOCK_STREAM â†’ socket.SOCK_DGRAM.
2. No connection setup: remove connect() â€” UDP is connectionless.
3. Sending: send()/sendall() â†’ sendto(data, (host, port)) â€” every datagram carries its destination.
4. Receiving: recv() â†’ recvfrom(bufsize) â€” returns (data, address) because the sender must be identified.
5. No listen()/accept() on the peer; a UDP "server" just bind()s and calls recvfrom().
6. Error handling: no automatic retransmission â€” timeouts and retries must be implemented by the application.

■ II. Example â€” UDP Client

• import socket
• s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
• s.sendto(b"Hello", ("localhost", 5000))
• data, addr = s.recvfrom(1024)
• print(data); s.close()